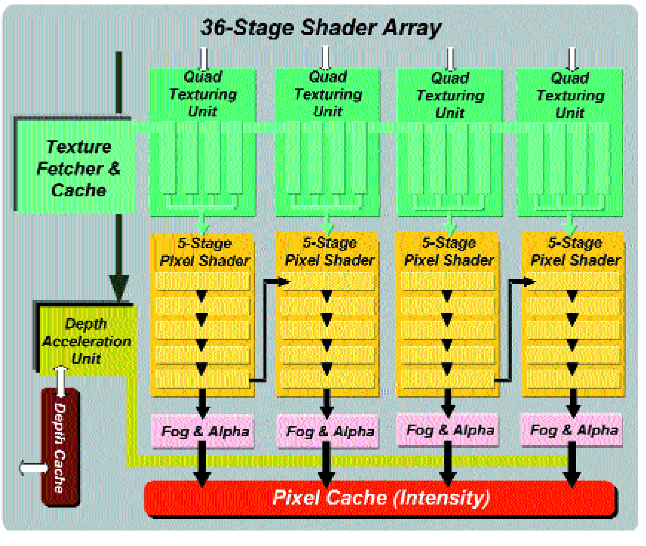
els even when executing complex vertex shader programs on high-resolution 3D models. Upcoming games such as *DOOM III* use more than one texture on the models within the game. These multiple textures give the models a more natural look, allowing an in-game car to have a metallic shine, or for a monster to sport a more believable snarl. This multiple-texturing will force cards such as the GeForce4 to take more than one pass to texture a particular model. The more the number of passes needed, the slower the game will run. Parhelia-512 is the first GPU to sport four vertex shader units, enabling the chip to process up to four textures in a single pass. This architecture can also be helpful in current games, which do not use four multiple textures on their models. For example, any game using the *Quake III Arena* engine uses not more than two textures for their models. In situations like these, the Parhelia can use its two idle shader units to perform anisotropic and trilinear filtering at virtually no performance hit.

The chip is not slack in the pixel shader department either. With four programmable texture and five programmable pixel shader stages on each of its four pixel pipelines, Parhelia's 36-stage Shader Array boasts the largest and most powerful pixel rendering pipeline to date. The array will allow the chip to handle up to 10 pixel shader operations across two pairs of pixel pipelines in a single pass if necessary, and although this has its drawbacks (in terms of loss of pixel fillrate), the efficiency gained will allow games to sport effects which are highly impractical as of now.



The Parhelia has four pixel pipelines, each with four quad-texture units. The vertex processing array has four vertex shading units coupled with an instruction cache. Hence, four textures can be used per clock. Each texture unit is then interfaced to a five-stage DirectX 8.1 pixel shader. The entire texture processing engine supplies data to the rest of the core via a 512-bit internal controller

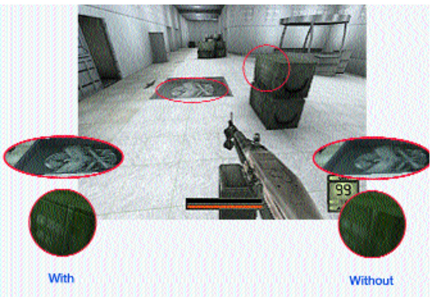
Visual volleys

Parhelia-512 also offers a host of other 3D scene and desktop enhancing features:

■ **10-bit GigaColor Technology:** GPUs have to deal with a lot of number-crunching—vertex shader operations, pixel shader operations, filtering, and antialiasing. The current 32-bit colour system gives each colour channel, viz. Red, Green, Blue and Alpha (used for transparency) an 8-bit value. This is not sufficient to accurately portray a pixel—since it goes through so many mathematical operations, there is bound to be some data loss. The results are artefacts like banding and discolouration. The Parhelia takes the first small step towards alleviating this problem. By re-distributing the 32-bits currently available, as 10-10-10-2 (RGBA), Matrox has allowed for a four-fold increase in accuracy per individual colour channel. As a result, the Parhelia is capable of simultaneously displaying over one billion colours. Full 10-bit colour precision is available for 2D, 3D, DVD and video.

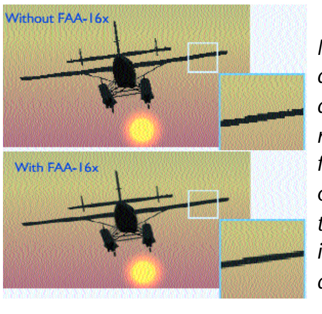
■ **64 Super Sample Texture Filtering:** Most video cards today have a standard pipeline structure—their four pixel pipelines each consist of two texture mapping units (TMU) to form a 4x2 structure. The Parhelia makes a stark departure from this norm and boasts of a 4x4 pixel pipe/TMU structure instead. This gives the chip an advantage while filtering textures—a process that gives a game a more clear and artefact-free look. While present cards are brought to their knees with trilinear or anisotropic filtering turned on, thanks to the power of its 4x4 pipeline, the Parhelia can yield 16 bilinear texels or eight trilinear texels or one anisotropic filtered texel per clock, providing a fast, efficient and beautiful solution.

The Parhelia has advanced texture filtering units which allow for the dynamic allocation of up to 64 texture samples per clock, which is double the number available on competing GPUs, resulting in improved visual quality.



The Parhelia's advanced texture filtering dramatically increases the level of details on textures. Notice the quality difference with this effect present and absent

■ **16x Fragment Antialiasing (FAA-16x):** Antialiasing refers to the process of removing the jagged edges of on-screen 3D objects and models. Current video cards employ a variant of the so-called full-screen antialiasing process (FSAA), where entire polygons are antialiased en masse. This method is highly inefficient and consumes a lot of memory bandwidth. The Parhelia follows a more conservative process: instead of antialiasing entire polygons, it identifies just the edge polygons and removes the jaggedness off their borders.



Maximum antialiasing quality with minimal performance overhead, all thanks to an intelligent approach

Matrox claims that edge pixels constitute only about 5 per cent of on-screen real estate per frame. Since it has to work on such a reduced number of pixels, the Parhelia can apply a very high 16x supersampling solution, without a significant performance hit. However, this process is not compatible with all games and it cannot remove artefacts from textures (like pixel-popping) that traditional FSAA solutions can.

■ **Hardware Displacement Mapping (HDM):** The goal behind HDM is to generate more realistic 3D terrain and characters. The process is quite similar to bump-mapping: keeping in mind the type